AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

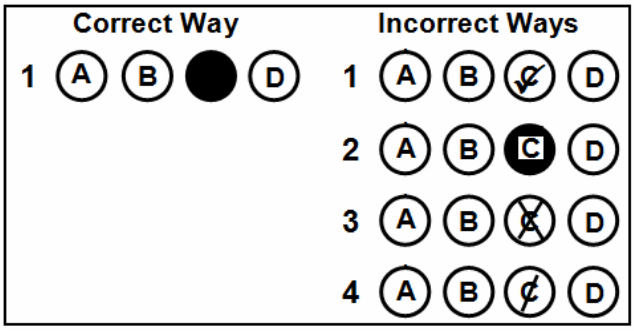
MAY 2014

Computer Science Paper I

Time: 35 minutes Marks: 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

1. Frequent use of computers may lead to all of the following problems EXCEPT

   A. blood pressure.
   B. eye strain.
   C. backache.
   D. RSI (Repetitive Strain Injury).

2. All of the following are the actions taken by antivirus software to prevent intrusive access to a computer's memory EXCEPT that it

   A. formats a disk.
   B. scans the memory.
   C. eliminates the virus.
   D. examines the files for the virus.

3. Which of the following is legal during internet access?

   A. Use each other's computers without permission
   B. Hide your personal information on web
   C. Download software using Torrent
   D. Spread computer viruses

4. Which of the following is a web mail application?

   A. Facebook
   B. Hotmail
   C. Skype
   D. Viber

5. Secure websites like 'google.com' and 'microsoft.com' are

   A. accessible via special hardware.
   B. accessible with permission.
   C. firewall protected.
   D. password protected.

6. In an effective problem solving approach, which of the following may reduce total programming time and cost?

   A. Analyzing the program
   B. Coding the program
   C. Documenting the program
   D. Testing the program

7. All of the following are the basic steps for writing an algorithm EXCEPT

   A. generating outputs.
   B. performing mathematical operations.
   C. performing debugging.
   D. supplying input.

8. Which of the following symbols is used to indicate 'start' and 'end' of a flowchart?

   A. 
   B. 
   C. 
   D. 

9. All of the following errors are examples of syntax errors EXCEPT

   A. A = X/0
   B. A = X - / 3
   C. A = 2(X + 6)
   D. $2A = X^2 - 4$

10. The purpose of DELETE statement in GW BASIC is to

    A. delete files from a disk.
    B. delete line numbers from a program.
    C. delete programs from memory.
    D. delete text from screen.

11. Representation of the algebraic expression $\frac{2xy}{2x}$ in GW BASIC is

A. $(2xy)/(2x)$
B. $(2*x*y)\backslash(2*x)$
C. $(2*x*y)/(2*x)$
D. $(2*xy)/(2*x)$

12. In GW BASIC, which of the following commands is used for displaying the program list on the printer?

A. LIST
B. LLIST
C. LPRINT
D. PRINT

13. Consider the following GW BASIC program.

```
10 CLS
20 FOR A = 1 TO 5
30 LET B  = B + A
40 END
```

Which of the following errors will be generated in this program?

A. FOR without STEP
B. FOR without RETURN
C. FOR without PRINT
D. FOR without NEXT

14. The concept of writing FOR loop within a WHILE loop is called

A. double loop.
B. infinite loop.
C. nested loop.
D. random loop.

15. Which of the following statements is valid about IF THEN ELSE?

A. It is used for repeating program statements several times.
B. It is used for unconditional branching.
C. It generates errors with GOTO statement.
D. It is used for making decisions regarding program flow.

16. Which of the following is used for reversing or inverting the logical condition in a GW BASIC program?

A. AND
B. OR
C. ON
D. NOT

17. Consider the following GW BASIC program.

```
10 Let A = 6
20 WHILE A >= 0
30 PRINT A
40 A = A – 2
50 WEND
60 END
```

How many times will this program be executed?

A. 7
B. 6
C. 5
D. 4

18. In GW BASIC, the CORRECT declaration of two dimensional arrays is

A. Array(3, 2) DIM
B. Array[3,2] DIM
C. DIM Array(3, 2)
D. DIM Array [3, 2]

19. Which of the following statements will return the value 1?

A. PRINT CINT(0.99)
B. PRINT FIX(0.99)
C. CINT(0.99)
D. RND(0.99)

20. Which of the following GW BASIC statements will produce a circle on the screen?

A. CIRCLE(160, 100, 50)
B. CIRCLE(160, 100), 50
C. DRAW(160, 100, 50)
D. DRAW(160, 100), 50

21. The correct syntax of writing Area = πR^2 in GW BASIC is

A. Area = 22/7 * R^2
B. Area = 22/7 $\times$ R^2
C. Area = 22/7 * R^ 2
D. Area = 22/7 $\times$ R^ 2

22. Which of the following is the CORRECT syntax for feeding string data through the keyboard?

A. INPUT "STRING", string
B. INPUT (STRING), string
C. INPUT "STRING" String$
D. INPUT "STRING", STRING$

23. Which of the following will return the remainder of an integer division?

A. ABS
B. INT
C. MOD
D. REM

24. According to the rules of Boolean algebra, $A+\overline{A}$ is equal to

A. 0
B. A
C. 1
D. $\overline{A}$

25. Using the duality principle, the complement of Boolean expression C + D will be

A. $C - D$
B. $C \cdot D$
C. $C - \overline{D}$
D. $C \cdot \overline{D}$

END OF PAPER